

DSA0503 – Query Processing for Data Science

CO2_AT2_Schema Analysis

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Experiment 1: Library Management Database

Aim

To use Python basics and the SQLite (sqlite3) library to connect to a database and retrieve the member name, book title, and issue date for all books issued after **1 January 2026**.

Requirement Analysis

A library maintains the following database schema:

- **Member (Member_ID, Member_Name, Phone)**
- **Book (Book_ID, Book_Title, Author)**
- **Issue (Issue_ID, Member_ID, Book_ID, Issue_Date)**

The objective is to retrieve:

- Member Name
- Book Title
- Issue Date

for all books issued after **01-01-2026**.

Database Schema

DB Browser for SQLite - C:\Users\HP\Downloads\library.db		
File Edit View Tools Help		
New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database		
Database Structure Browse Data Edit Pragmas Execute SQL		
Create Table Create Index Modify Table Delete Table Print Refresh		
Name	Type	Schema
Tables (3)		
Book		CREATE TABLE Book(Book_ID INTEGER PRIMARY KEY, Book_Title TEXT, Author TEXT)
Book_ID	INTEGER	"Book_ID" INTEGER
Book_Title	TEXT	"Book_Title" TEXT
Author	TEXT	"Author" TEXT
Issue		CREATE TABLE Issue(Issue_ID INTEGER PRIMARY KEY, Member_ID INTEGER, Book_ID INTEGER, Issue_Date TEXT)
Issue_ID	INTEGER	"Issue_ID" INTEGER
Member_ID	INTEGER	"Member_ID" INTEGER
Book_ID	INTEGER	"Book_ID" INTEGER
Issue_Date	TEXT	"Issue_Date" TEXT
Member		CREATE TABLE Member(Member_ID INTEGER PRIMARY KEY, Member_Name TEXT, Phone TEXT)
Member_ID	INTEGER	"Member_ID" INTEGER
Member_Name	TEXT	"Member_Name" TEXT
Phone	TEXT	"Phone" TEXT
Indices (0)		
Views (0)		
Triggers (0)		

SQL Query

```
SELECT Member.Member_Name,
       Book.Book_Title,
       Issue.Issue_Date
FROM Issue
JOIN Member
ON Issue.Member_ID = Member.Member_ID
JOIN Book
ON Issue.Book_ID = Book.Book_ID
WHERE Issue.Issue_Date > '2026-01-01';
```

Python Program

```
import sqlite3

# Connect to the correct database
conn = sqlite3.connect(r"C:\Users\HP\Downloads\library.db")

cursor = conn.cursor()

query = """
SELECT Member.Member_Name,
       Book.Book_Title,
       Issue.Issue_Date
FROM Issue
JOIN Member
ON Issue.Member_ID = Member.Member_ID
JOIN Book
ON Issue.Book_ID = Book.Book_ID
WHERE Issue.Issue_Date > '2026-01-01';
"""

cursor.execute(query)

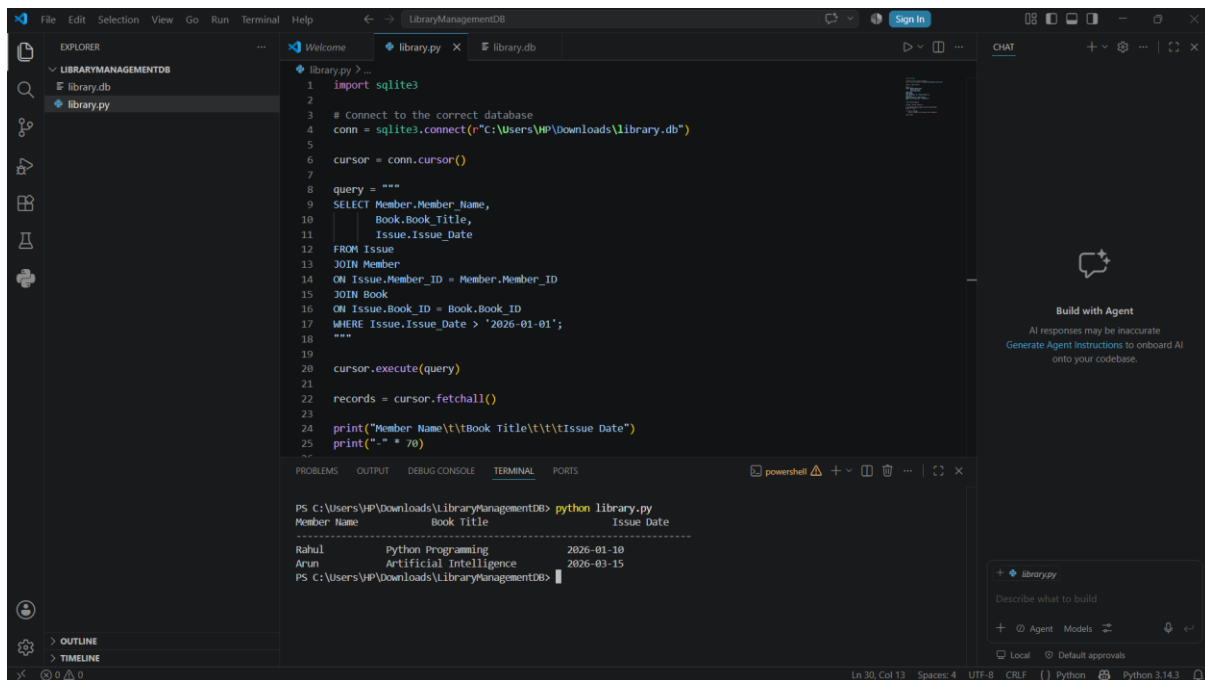
records = cursor.fetchall()

print("Member Name\t\tBook Title\t\t\tIssue Date")
print("-" * 70)

for row in records:
    print(f"{row[0]:<15}\t{row[1]:<30}\t{row[2]}")

conn.close()
```

Output



The screenshot shows a Visual Studio Code editor with a Python file named `library.py` open. The script connects to a SQLite database, executes a SQL query, and prints the results. The terminal output shows the command `python library.py` being executed and the resulting data.

```
1 import sqlite3
2
3 # Connect to the correct database
4 conn = sqlite3.connect(r"C:\Users\HP\Downloads\library.db")
5
6 cursor = conn.cursor()
7
8 query = """
9 SELECT Member.Member_Name,
10        Book.Book_Title,
11        Issue.Issue_Date
12 FROM Issue
13 JOIN Member
14 ON Issue.Member_ID = Member.Member_ID
15 JOIN Book
16 ON Issue.Book_ID = Book.Book_ID
17 WHERE Issue.Issue_Date > '2026-01-01';
18 """
19
20 cursor.execute(query)
21
22 records = cursor.fetchall()
23
24 print("Member Name\t\tBook Title\t\tIssue Date")
25 print("-" * 70)
26
```

Terminal Output:

```
PS C:\Users\HP\Downloads\LibraryManagementDB> python library.py
Member Name      Book Title      Issue Date
-----
Rahul            Python Programming 2026-01-10
Arun             Artificial Intelligence 2026-03-15
PS C:\Users\HP\Downloads\LibraryManagementDB>
```

Result

The SQLite database was successfully connected using Python (sqlite3). The required records were retrieved by joining the **Member**, **Book**, and **Issue** tables. The program displayed the member name, book title, and issue date for books issued after **1 January 2026**.

Conclusion

The experiment was completed successfully using Python and SQLite. Database connectivity was established, SQL JOIN operations were performed, and the required records were retrieved based on the specified date condition.

Experiment 2: Hotel Reservation Database

Aim

To use Python basics and the SQLite (`sqlite3`) library to connect to a database and retrieve the customer name, room type, and check-in date for customers who have reserved **Deluxe** rooms.

Requirement Analysis

A hotel maintains the following database schema:

- **Customer (Customer_ID, Customer_Name, City)**
- **Room (Room_ID, Room_Type, Rent)**
- **Reservation (Reservation_ID, Customer_ID, Room_ID, Check_In_Date, Check_Out_Date)**

The objective is to retrieve:

- Customer Name
- Room Type
- Check-In Date

for customers who have reserved **Deluxe** rooms.

Database Schema

DB Browser for SQLite - C:\Users\HP\Downloads\hotel.db		
File Edit View Tools Help		
New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database		
Database Structure Browse Data Edit Pragmas Execute SQL		
Create Table Create Index Modify Table Delete Table Print Refresh		
Name	Type	Schema
Tables (3)		
Customer		CREATE TABLE Customer(Customer_ID INTEGER PRIMARY KEY, Customer_Name TEXT, City TEXT)
Customer_ID	INTEGER	"Customer_ID" INTEGER
Customer_Name	TEXT	"Customer_Name" TEXT
City	TEXT	"City" TEXT
Reservation		CREATE TABLE Reservation(Reservation_ID INTEGER PRIMARY KEY, Customer_ID INTEGER, Room_ID INTEGER, CI
Reservation_ID	INTEGER	"Reservation_ID" INTEGER
Customer_ID	INTEGER	"Customer_ID" INTEGER
Room_ID	INTEGER	"Room_ID" INTEGER
Check_In_Date	TEXT	"Check_In_Date" TEXT
Check_Out_Date	TEXT	"Check_Out_Date" TEXT
Room		CREATE TABLE Room(Room_ID INTEGER PRIMARY KEY, Room_Type TEXT, Rent REAL)
Room_ID	INTEGER	"Room_ID" INTEGER
Room_Type	TEXT	"Room_Type" TEXT
Rent	REAL	"Rent" REAL
Indices (0)		
Views (0)		
Triggers (0)		

SQL Query

```
SELECT Customer.Customer_Name,
       Room.Room_Type,
       Reservation.Check_In_Date
FROM Reservation
JOIN Customer
ON Reservation.Customer_ID = Customer.Customer_ID
JOIN Room
ON Reservation.Room_ID = Room.Room_ID
WHERE Room.Room_Type = 'Deluxe';
```

Python Program

```
import sqlite3

# Connect to SQLite database
conn = sqlite3.connect(r"C:\Users\HP\Downloads\hotel.db")

cursor = conn.cursor()

query = """
SELECT Customer.Customer_Name,
       Room.Room_Type,
       Reservation.Check_In_Date
FROM Reservation
JOIN Customer
ON Reservation.Customer_ID = Customer.Customer_ID
JOIN Room
ON Reservation.Room_ID = Room.Room_ID
WHERE Room.Room_Type='Deluxe';
"""

cursor.execute(query)

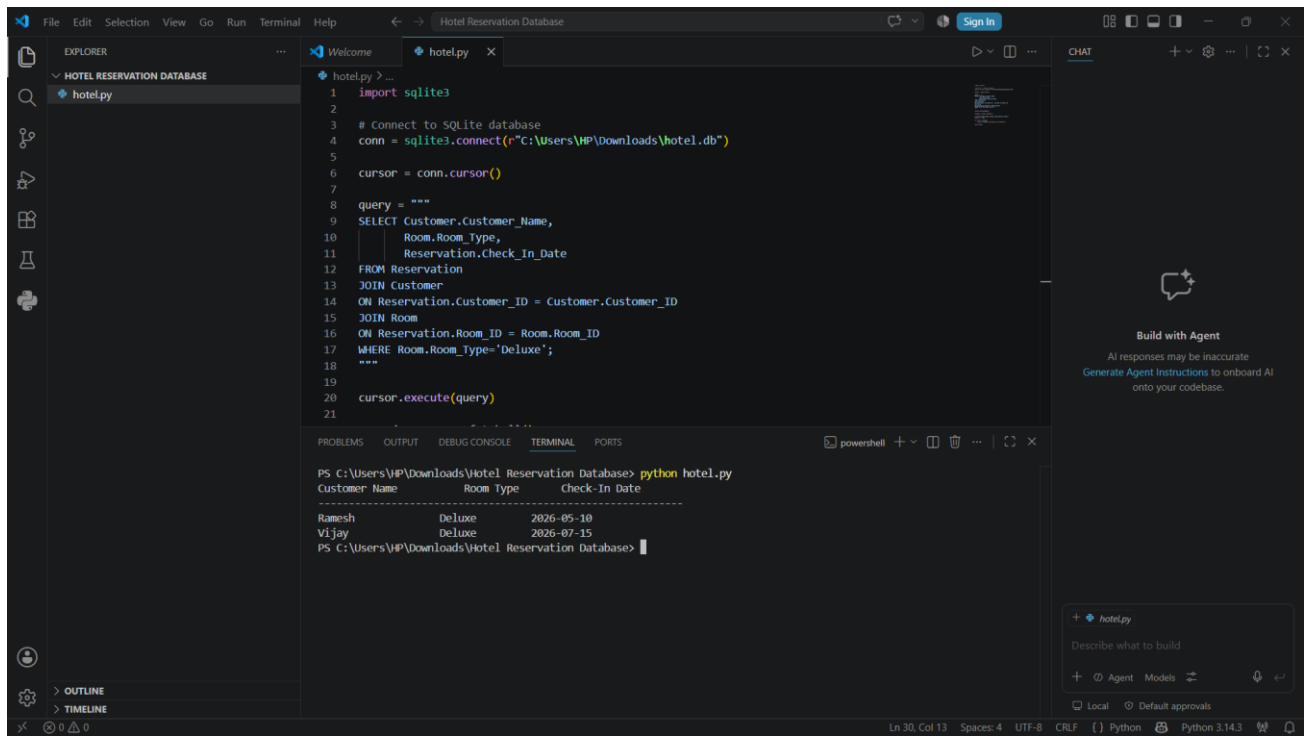
records = cursor.fetchall()

print("Customer Name\t\tRoom Type\tCheck-In Date")
print("-" * 60)

for row in records:
    print(f"{row[0]:<20}{row[1]:<15}{row[2]}")

conn.close()
```

Output



The screenshot shows a Visual Studio Code editor with a Python file named `hotel.py` open. The script connects to a SQLite database at `C:\Users\HP\Downloads\hotel.db` and executes a SQL query that joins the `Customer`, `Room`, and `Reservation` tables, filtering for `Deluxe` rooms. The terminal output shows the results of the query:

```
PS C:\Users\HP\Downloads\Hotel Reservation Database> python hotel.py
Customer Name      Room Type      Check-In Date
-----
Ramesh             Deluxe         2026-05-10
Vijay              Deluxe         2026-07-15
PS C:\Users\HP\Downloads\Hotel Reservation Database>
```

Result

The SQLite database was successfully connected using Python (`sqlite3`). The required records were retrieved by joining the **Customer**, **Room**, and **Reservation** tables. The program displayed the customer name, room type, and check-in date for customers who reserved **Deluxe** rooms.

Conclusion

The experiment was completed successfully using Python and SQLite. Database connectivity was established, SQL `JOIN` operations were performed, and the required records were retrieved based on the room type **Deluxe**.